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4 April 1960
SUPERSEDING
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16 May 1955

MILITARY SPECIFICATION

RELEASES, LIFESAVING EQUIPMENT, HYDRAULIC AND MANUAL

1. SCOPE

1.1 This specification covers manual and automatic hydraulic release devices for freeing CO_2 inflatable life boats and life floats in sinking vessels.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood-Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Fiber.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-B-10347 - Boxes: Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

FEDERAL
Fed. Std. No. 151 - Metals; Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS
S8201-495181 - Arrangement and Equipment for Testing Hydraulic Release Devices.
S8201-921889 - Boat Stowage, CO_2 Inflatable Lifeboat (15 Person) Single and Multiple Types, Details.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material. - The material used in the construction of hydraulic release-devices shall be corrosion-resisting steel for all metallic parts. The material shall be free from any defects which may affect the operation or serviceability.

3.2 Design and construction. -

3.2.1 The hydraulic release-devices shall be designed for releasing life rafts or buoyant containers carrying water, food, medical supplies, navigational instruments, and similar items, upon submergence of the vessel due to sinking, and shall be as simple and light as is commensurate with reliability and good engineering practice. The surfaces of the parts bearing on the release pin shall be made smooth with slightly rounded edges.

3.2.2 Hydraulic release-devices shall consist of two distinct, separable parts which can be introduced into the lashing of life-saving appliances in such a manner that separation of the parts will also separate the lashing and allow the appliances to be free for removal. In no case shall the physical dimensions of the release device exceed 6 inches by 12 inches.

3.2.3 There shall be two means of separating the two parts of the device, either one of which shall fully effect a separation of the lashing. These methods shall be as follows:

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(a) Manually. - The device shall release by any manual operation without it being necessary to employ a wrench, knife, or tool of any kind. It is intended that the manual release be used provided the ship does not sink too rapidly to preclude such a procedure. If a lever release is used, open and shut positions shall be distinctly marked on the manual release. Manual release shall be accomplished readily without undue expenditure of force and time.

(b) Automatically. - Due to the force of hydrostatic pressure of the water acting upon a diaphragm to withdraw a pin, it is intended that the automatic release come into play in the event the vessel sinks too rapidly to allow the manual release to be employed.

3.2.4 The design shall be such that extreme changes in temperature, ranging from minus 30° to plus 120° Fahrenheit (F) will not cause the device to trip or otherwise be inoperative. The design shall also be such that there is a minimum possibility of icing conditions or dirt preventing access of water to the diaphragm or other mechanism actuating the release-device.

3.2.5 Hydraulic release devices shall automatically release lashings which restrain buoyancy loads of 400, 800 and 1200 pounds for the 800 pound capacity device and buoyancy loads of 1250, 2500 and 3750 pounds for the 2500 pound capacity device at a depth of 25 ±15 feet. A single device designed to automatically release lashings, which restrain buoyancy loads of 400 to 3750 pounds, at a depth of 25 ±15 feet may be provided as an alternate for either the 800 or the 2500 pound capacity device (see 6.1).

3.2.6 A single 3/4 inch diameter hole shall be provided at each end of the release device for lashing purposes.

3.2.7 The hydraulic mechanism shall be so designed that a releasing force of not less than 5 pounds is developed for each 10 feet the device is submerged.

3.2.8 The diaphragm shall be suitably shielded to prevent damage or accidental injury due to tampering or to the impact of heavy seas. A protecting cover or guard shall be fitted for the automatic release pin to protect its operation from paint, dirt, salt spray or other deleterious deposits.

3.2.9 Replacement diaphragms shall be provided in quantities totaling 10 percent the total number of the releases installed on each ship. The device shall be so designed that replacement diaphragms may be installed by the ship's personnel.

3.3 Marking. - The design capacity, the method of manual release, and the notation "DO NOT PAINT" shall be stamped on a plainly visible portion of the exterior of the releasing device. The design capacity to be marked on the device shall be 800 pound, 2500 pound, or 800 to 2500 pound, as applicable.

3.4 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - For purposes of sampling, a lot shall consist of not more than 500 hydraulic release devices of the same design and size offered for inspection and tests.

4.2.2 Sampling for visual and dimensional examination. - A random sample of release devices shall be selected from each lot of material offered for examination of visual and dimensional characteristics with lot acceptance based in accordance with table I and Standard MIL-STD-105.

Table I - Sampling for visual and dimensional examination
AQL (approx.) = 1.0 percent defective.

Number of release devices in inspection lot	Number of release devices in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	all	-	-
16 to 25	15	0	1
26 to 40	25	0	1
41 to 110	35	1	2
111 to 180	50	1	2
181 to 300	75	2	3
301 to 500	110	3	4

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Table II - Sampling for lot acceptance tests

Number of release devices in inspection lot	Number of release devices in sample	Acceptance number (failures in the tests)	Rejection number (failures in the tests)
15 and under	4	0	1
16 to 25	5	0	1
26 to 40	7	0	1
41 to 110	10	0	1
111 to 180	12	0	1
181 to 300	16	0	1
301 to 500	20	0	1

4.2.3 Sampling for lot acceptance tests (at a laboratory). - From each lot the Government inspector shall select at random release devices in accordance with table II, and shall forward the samples to a laboratory satisfactory to the bureau or agency concerned for the tests specified 4.4.

4.3 Visual and dimensional examination. - Each of the sample release devices selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any release device in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective release devices in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for inspection provided the contractor has removed all nonconforming release devices. The inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

4.4 Lot acceptance tests. - Each of the sample release devices selected in accordance with 4.2.3 shall be submitted to the following tests in the order listed to determine that the release devices conform to this specification.

Test	Paragraph reference
Temperature	4.5.3
Submergence	4.5.1
Corrosion resisting	4.5.2

If any release device in the sample fails to pass any of the tests, the lot represented by the sample shall be rejected. Rejected lots may be offered again for tests provided the contractor has removed all nonconforming release devices.

4.5 Methods of test. -

4.5.1 Submergence test. - The sample shall be tested by applying loads as indicated in 3.2.5 by means of actual weighted or applied tension and submerging in water or in a pressure test cylinder. The sample shall automatically trip and release the load at a depth of 25 ± 15 feet or at the equivalent water pressure in order to determine compliance with 3.2.5 (see Drawing S8201-495181).

4.5.2 Corrosion-resistance test. - After the completion of the submergence test specified in 4.5.1, the sample shall be exposed to a 20 percent salt spray test for 160 continuous hours in accordance with Standard Fed. Std. No. 151. At the conclusion of the above test, the sample shall be entirely serviceable and shall show only a minimum amount of corrosion.

4.5.3 Temperature test. - The sample in the condition "as received" shall be placed in a cold box at minus 30° F. for 4 hours. Upon completion of this test the device shall be opened for inspection and shall show no significant change in the position of the hydraulic or manual control as a result of the low temperature exposure. This test shall be repeated after the submergence and the corrosion-resistance tests (see 4.5.1 and 4.5.2).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1). -

5.1.1 Level A. - Release-devices shall be individually preserved and packaged in accordance with Method III of Specification MIL-P-116.

5.1.1.1 Unit containers. - Unit containers shall conform to Specification PPP-B-566, PPP-B-676, PPP-B-665 at the option of the contractor.

5.1.1.2 Intermediate containers. - Unit packages in an intermediate container shall be as specified (see 6.1). Intermediate containers shall conform to Specification PPP-B-636. The gross weight of fiber boxes shall not exceed 20 pounds.

5.1.2 Level C. - Releases-devices shall be preserved and packaged in accordance with the supplier's commercial practice.

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5.2 Packing (see 6.1).-

5.2.1 Level A.- Release-devices, packaged as specified (see 6.1), shall be packed in overseas type, wood cleated fiberboard, nailed wood, fiber, wirebound wood, wood cleated veneer paper overlaid, or wood cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-636 Class 2 or 3, PPP-B-585 (class 3), MIL-B-10377 or PPP-B-601, respectively at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch wide tape conforming to Specification PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes may be banded with tape conforming to type IV, Specification PPP-T-97 and the appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds: fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes conforming to Class 2 or 3 of Specification PPP-B-636, closed, sealed and banded as specified herein, and used as the shipping container need not be overpacked.

5.2.2 Level B.- Release-devices, packaged as specified (see 6.1) shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood cleated veneer paper overlaid boxes or class 1 fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, (class 1), PPP-B-601, MIL-B-10377 or PPP-B-636, respectively at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes conforming to Specification PPP-B-636, closed as specified herein, and used as the shipping container need not be overpacked.

5.2.3 Level C.- Release-devices, packaged as specified (see 6.1), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order or herein, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Capacity of devices required (see 3.2.5).
(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
(d) Unit packages in an intermediate container (see 5.1.1.2),

6.2 The 800 pound capacity release-device shall be used with each 10, 15, 25 and 40 person life float stowage and each single CO_2 inflatable lifeboat stowage and each lashed buoyant net stowage. The 2500 pound capacity release-device shall be used with each 60 person life float and with each stowage of two or more CO_2 inflatable lifeboats or life floats.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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